

Introduction, Methods, and Results

Working manuscript preview

Introduction

Molecular conformer generators provide three-dimensional structures for shape comparison, pharmacophore screening, and structure-based molecular modeling. For these applications, an ensemble is useful when it contains conformations close to those a molecule adopts in the relevant experimental setting. Previous work has examined how ensemble size, diversity, and energy minimization affect recovery of bioactive conformations and downstream screening performance.[1] These observations motivate an application-specific evaluation of generated ensembles: how reliably does a method recover an experimentally observed ligand geometry, and how much sampling is required to do so?

Conformer generators approach this problem through different sampling procedures. Distance-geometry methods combine molecular connectivity with geometric constraints and experimental torsional preferences.[2] Learned approaches include diffusion over torsional degrees of freedom, continuous conformer fields, and models that combine molecular language representations with coordinate generation.[3–5] Low-energy conformer generation provides another objective, represented here by LoQI and its associated computed ensembles.[6] These methods offer different ways to construct an ensemble, but their relative performance must be established for the intended reference structures and sampling regime.

Several properties of an ensemble need to be distinguished in that comparison. Physical-validity checks assess molecular identity and local geometric plausibility.[7] Geometric diversity describes the spread of generated structures, whereas recovery measures whether that spread includes a particular experimental conformation. An ensemble can contain many distinct conformers and still miss the target. Conversely, a large ensemble can recover the target through a small fraction of its samples, even when its typical conformer is relatively distant. These measurements do not, by themselves, establish binding affinity or an equilibrium conformational population.

We therefore evaluate conformer generators from the perspective of experimental-conformation recovery, interpreting their ranking jointly with sampling budget, geometric diversity, and molecular complexity. The primary comparison uses 94 CASF-associated ligand entries matched to ChEMBL3D, with a larger matched collection used to examine the consistency and scope of the findings. We compare classical and learned generators, including an autoregressive Qwen 1.7B model, at a common candidate target and at smaller sampling targets. A specified RMSD cutoff provides the principal comparison; threshold curves show how that comparison changes with the required structural agreement. Diversity is assessed across clustering radii, and recovery is stratified by molecular size and flexibility.

A separate panel of 23 drug molecules extends the question from recovering one experimental shape to covering multiple observed conformations. In this setting, reference coverage and the fraction of generated samples near a reference describe complementary properties of an ensemble. Together, the two evaluations distinguish recovery by the best member of a sampled pool from the concentration of that pool near observed structures. The intended contribution is a comparative account of which generators serve these objectives under the evaluated conditions, including where rankings change with sampling, structural tolerance, and molecular complexity.

Methods

Benchmark datasets and molecular identity

We evaluated conformer ensembles against experimental ligand geometries using two complementary settings: recovery of one bound conformation per protein–ligand complex and coverage of several observed conformations of the same molecule. The first setting used the project’s CASF-2016 core and larger reference ligand collections, denoted core and ref.[8] Core was the primary comparison; ref provided a larger supporting cohort. Ligands were matched to ChEMBL3D by exact canonical, heavy-atom isomeric SMILES. ChEMBL3D provided molecular topologies and a separate computed conformer ensemble for comparison.[6] Entries were retained when the ligand could be parsed, a matching topology was available, and at least

one eligible torsion was identified by the SMARTS pattern `[!$(***)&!D1]-!@[!$(***)&!D1]`. This torsion eligibility rule differs from the rotatable-bond descriptor used to stratify the results.

We resolved stereoisomers within each ChEMBL3D parent record using isomeric SMILES and the tetrahedral and double-bond stereochemistry layers of InChI. The selected topology record was retained in the mapping, and stored coordinates were filtered to the matching stereoisomer. Thus, both the reference ensemble and its conformer count were specific to the mapped stereoisomer. The resulting core cohort contained 94 complex entries and 94 ChEMBL parent compounds; ref contained 1,236 entries and 1,042 parent compounds. The cohorts shared no complex identifiers but included 17 common parent compounds. We retained separate complex entries because they provide separate experimental reference geometries, while accounting for repeated compounds in the uncertainty analysis.

The second setting used a curated set of 23 druglike molecules with 2,450 stored experimental reference conformations in total, ranging from 23 to 716 per molecule. We used the supplied molecular identities and reference coordinates without treating the frequency of a structure in this collection as an equilibrium population. These references represent observed shapes; they do not exhaust the conformations that a molecule can adopt.

Conformer generators

The main comparison included four classical generators, five external learned generators, and one representative Qwen checkpoint. The classical generators comprised RDKit distance geometry and torsion perturbation, each with and without energy minimization. RDKit conformers were generated with ETKDGv3 from a topology-only molecule, using random starting coordinates, enforced chirality, and no RMSD pruning.[2] The minimized variant applied MMFF94s for up to 500 iterations. Structures that reached the iteration limit were retained if the optimization returned normally; convergence was not required. The two variants used independently sampled pools.

Torsion perturbation started from the matched ChEMBL3D topology’s stored geometry. In each trial, every eligible torsion received an independent uniform angular displacement between -120° and 120° . Candidates were rejected if a nonbonded heavy-atom distance was below 0.7 times its distance-geometry lower bound; bonded and 1,3 pairs were excluded from this check. The minimized variant additionally applied MMFF94s and repeated the clash check. Candidate generation continued toward the requested pool size before the common validity filter was applied.

The external methods were LoQI,[6] Torsional Diffusion,[3] Molecular Conformer Fields (MCF, drugs-L),[4] NExt-Mol (DMT-L),[5] and FLOWR.root, referred to here as FlowR.[9] We evaluated their saved conformer outputs through the benchmark adapters. FlowR used the ligand-only v2.2 checkpoint with the input atom and bond graph fixed during coordinate generation. Its adapter included up to 200 steps of force-field cleanup, using MMFF with UFF as a fallback, and stereochemistry checks; mirror reflection was allowed when it restored the requested stereoisomer. Remaining stereochemical mismatches were rejected. Consequently, an ensemble labeled raw in the benchmark can already include preprocessing specific to its generator.

The Qwen generator produced conformations autoregressively from molecular SMILES. The main analysis used the 1.7-billion-parameter model with finite scalar quantization (FSQ), evaluated at pretraining step 47,023. The multi-reference comparison used the same representative Qwen checkpoint and the five external methods. Training-set overlap with the benchmark molecules has not been established by a common audit.

Ensemble sizes and validity filtering

The main CASF comparison used a target of 1,000 conformers per ligand before the common validity filter. A smaller count tier was sampled without replacement from each available candidate pool, with a target equal to the stereoisomer-specific ChEMBL3D conformer count. Subsets were capped at the available pool size and then subjected to the same validity filter. The classical-generation launcher used base seed 42; the common saved-pool materializer used base seed 1729, with molecule- and method-specific seeds derived from these values.

We applied PoseBusters to assess molecular identity and intramolecular plausibility.[7] The configured checks covered loading, sanitization, InChI conversion, connectivity, radicals, molecular formula, bond con-

nectivity, tetrahedral and double-bond stereochemistry, bond lengths and angles, internal clashes, planarity, and energy ratio. The distance-geometry tolerances were 0.25 for bond lengths and angles and 0.3 for clashes, with hydrogens excluded from these distance checks. Planarity tolerance was 0.25 Å for the specified aromatic rings and trigonal carbon-carbon double-bond groups. The energy-ratio threshold was 100, using a reference ensemble of 50 conformations. No protein-contact checks were included.

A conformer passed only if all configured Boolean checks passed. Generated CASF ensembles were checked against the matched ChEMBL3D molecular reference; the stored ChEMBL3D comparison ensembles were checked against the CASF ligand. The same pass mask was applied to the full candidate pool and its sampled subsets. Rejected conformers were not replaced after filtering. Thus, the 1,000-conformer target does not imply equal retained ensemble sizes or equal numbers of generation attempts. We report retained counts alongside recovery. ChEMBL3D was evaluated as a finite stored ensemble, with a 2,000-conformer cap for the designated large reference entry, 1t1p, when necessary.

Conformational recovery and diversity

CASF recovery was evaluated on conformers that passed PoseBusters, using the experimental ligand geometry as the target. We removed hydrogens and used RDKit’s symmetry-aware `GetBestRMS` alignment. If this alignment raised an exception, the CASF implementation attempted `AlignMol` with an atom-index correspondence. An unresolved, nonfinite RMSD caused analysis of that ensemble to fail rather than silently removing the conformer.

For each ligand, we calculated the minimum and median RMSD over the retained ensemble. $\text{Hit}@ \delta$ was one when at least one conformer had $\text{RMSD} \leq \delta$, and zero otherwise, at thresholds of 0.25, 0.5, 0.75, and 2.0 Å. We used 0.75 Å as the principal recovery cutoff and the remaining thresholds to assess sensitivity. Cohort hit rates used all mapped entries as the denominator, including empty or missing outputs as misses. Mean minimum RMSD, mean within-ensemble median RMSD, and diversity summaries were calculated only over entries with defined values. These conditional summaries were reported separately from the full-cohort hit rates.

We measured geometric diversity by greedy clustering of the retained conformers at a principal heavy-atom RMSD threshold of 1.0 Å, with additional radii of 0.5, 2.0, and 3.0 Å for sensitivity analysis. Conformers were visited in stored order and assigned to the nearest existing cluster representative if their RMSD was below the threshold; otherwise, a new cluster was created. We reported the mean number of clusters across measurable ensembles. This measure describes geometric spread and depends on ensemble size and clustering order.

Coverage of multiple experimental conformations

For each druglike molecule, we compared the supplied generated pool with all of its experimental references using aligned heavy-atom RMSD. Recall coverage (COV-R) was the fraction of reference conformations with a generated neighbor below 0.75 Å. Precision coverage (COV-P) was the fraction of generated conformations with a reference neighbor below the same threshold. Matching distances MAT-R and MAT-P were the mean nearest-neighbor RMSDs in the reference-to-generated and generated-to-reference directions, respectively. Unlike CASF $\text{Hit}@ \delta$, coverage used a strict inequality. We averaged each metric equally across molecules. COV-P measures proximity to the available observations; a low value does not establish that the remaining conformers are physically inaccessible.

These coverage calculations used the supplied generated pools without a common PoseBusters filter. Validity was assessed separately. The imported Qwen evaluator counted reference or generated conformations with no finite RMSD match as coverage misses, whereas the local external-method evaluator excluded them from the corresponding coverage denominator. Both omitted undefined nearest-neighbor distances from MAT. We therefore treat the current multi-reference comparison as descriptive pending evaluation under a shared failure policy.

Statistical analysis

We calculated paired differences in recovery and coverage on the same complex entries or druglike molecules. For CASE, 95% confidence intervals were obtained from 3,000 bootstrap resamples of ChEMBL parent compounds, retaining all complex entries belonging to each sampled compound. Each resample used the mean over its complex entries, preserving the entry-weighted estimate while accounting for repeated compounds. Missing hits were assigned zero. For druglike method differences, we used 10,000 paired resamples of the 23 molecules. Figure 4A instead reports intervals for each method’s mean COV-R and COV-P, calculated from 10,000 molecule resamples.

Intervals were the 2.5th and 97.5th percentiles of the bootstrap distribution, using a NumPy random generator initialized with seed 20260924. These intervals were calculated from the molecule-level results for this analysis; they describe resampling uncertainty across the evaluated molecules, not variation across independent generation or training runs.

As an overlap sensitivity analysis, we repeated ref comparisons after excluding the 17 parent compounds also present in core, leaving 1,189 complex entries from 1,025 compounds. Exploratory size analyses used heavy-atom counts from the hydrogen-stripped experimental ligand, grouped as fewer than 20, 20–29, 30–39, and at least 40 atoms. Flexibility analyses used RDKit rotatable-bond counts from that same reference, grouped as 0–3, 4–6, 7–8, and at least 9. The reference-derived descriptors assigned each entry to the same group for every generator. Stratum hit rates included all mapped entries in the group, with missing outputs counted as misses. These descriptive analyses did not adjust size effects for flexibility or chemical class. Confidence intervals were not adjusted for multiple comparisons or checkpoint selection.

Results

Experimental-conformation recovery varied with structural tolerance

We compared generators on the 94-ligand core cohort by counting a ligand as recovered when at least one PoseBusters-passing conformer lay within 0.75 Å of its experimental geometry. Generated ensembles were evaluated at a target of 1,000 candidates before filtering, while ChEMBL3D-PB provided a comparison with the available stored conformers. Recovery percentages therefore describe the fraction of the 94 ligands for which the ensemble contained a matching conformer, with missing outputs counted as failures.

At this cutoff, the pretrained Qwen 1.7B FSQ model recovered 86 of 94 ligands (91.5%), compared with 84 for Torsional Diffusion (89.4%) and 81 for LoQI (86.2%). Recovery was lower for RDKit without minimization and ChEMBL3D-PB, which recovered 76 (80.9%) and 74 (78.7%) ligands, respectively (Table 1). Qwen recovered every ligand recovered by ChEMBL3D-PB and 12 additional ligands, giving a paired difference of 12.8 percentage points (95% confidence interval, 6.4–20.2). This gain was achieved with a substantially larger ensemble, however: Qwen retained 972.6 conformers per ligand on average, compared with 81.2 in the stored ChEMBL3D-PB ensembles.

Requiring a closer structural match changed the ordering between Qwen and LoQI, as shown by the threshold curves in Figure 1A. At 0.5 Å, LoQI recovered 83.0% of core ligands and Qwen recovered 78.7%; at 0.25 Å, the corresponding values were 55.3% and 53.2%. Qwen’s higher recovery at the principal cutoff thus did not imply higher recovery at every structural tolerance.

The larger ref cohort supported the high-budget result at 0.75 Å, with Qwen recovering 92.5% of 1,236 entries, compared with 86.8% for LoQI and 78.7% for ChEMBL3D-PB. The paired advantage over LoQI was 5.7 percentage points (95% confidence interval, 3.8–7.7), while excluding compounds shared with core preserved the advantage over ChEMBL3D-PB (14.0 points; 11.7–16.4). The latter comparison indicates that overlap between the two evaluation cohorts did not explain this result, although it does not address overlap with model training data.

The improvement in recovery was concentrated in the best members of the sampled ensembles. On core, Qwen’s mean minimum RMSD was 0.307 Å, compared with 0.339 Å for LoQI and 0.525 Å for ChEMBL3D-PB, whereas their mean within-ensemble median RMSDs were similar at 1.318, 1.305, and 1.326 Å, respectively. The higher recovery therefore did not imply that a typical Qwen conformer was closer to the experimental target.

Table 1: Recovery at the 1,000-candidate target after PoseBusters filtering. ChEMBL3D-PB is the finite stored comparison ensemble. Hit rates use all 94 core or 1,236 ref entries at 0.75 Å. Core N is the mean retained count over all entries; core RMSD is the mean minimum RMSD over measurable ensembles.

Method	Core hit (%)	Core N	Core RMSD (Å)	Ref hit (%)
ChEMBL3D-PB	78.7	81.2	0.525	78.7
RDKit raw	80.9	999.9	0.437	81.5
RDKit minimized	78.7	1000.0	0.512	77.3
Torsion raw	78.7	896.5	0.477	77.7
Torsion minimized	79.8	1000.0	0.535	77.1
LoQI	86.2	999.3	0.339	86.8
Torsional Diffusion	89.4	881.4	0.371	86.2
MCF drugs-L	86.2	866.8	0.366	84.6
NExT-Mol DMT-L	86.2	871.9	0.386	84.8
FlowR	86.2	920.0	0.414	85.5
Qwen 1.7B FSQ	91.5	972.6	0.307	92.5

Core RMSD denominators are 93 for MCF and NExT-Mol and 94 for the other listed methods. Missing recovery outcomes count as misses. The label *raw* may include generator-specific preprocessing (Methods).

Smaller sampling targets changed the comparison

Because the fixed-target comparison involved different retained ensemble sizes, we next reduced each generated candidate pool to the stereoisomer-specific ChEMBL3D count before applying the validity filter (Figure 1B). On core, Qwen recovery fell from 91.5% to 80.9%, with a mean of 78.3 retained conformers, bringing it close to ChEMBL3D-PB at 78.7% recovery and 81.2 conformers. The remaining paired difference was two ligands (2.1 percentage points; 95% confidence interval, -4.3 to 8.5), so the larger-target advantage was not resolved at this smaller sampling target.

On ref, reducing the target reversed the ordering: Qwen recovered 76.3% of entries with a mean of 63.3 retained conformers, compared with 78.7% and 66.2 for ChEMBL3D-PB. The paired difference of -2.4 percentage points (95% confidence interval, -4.7 to -0.1) shows that better recovery at the larger target did not translate into better recovery from a compact ensemble. These comparisons establish a dependence on sampling regime, but unequal retained counts prevent identification of a crossover at a particular valid-conformer count.

Geometric diversity did not reproduce the recovery ranking

To examine whether broader geometric sampling accounted for the recovery ordering, we compared the number of conformer clusters at a 1.0 Å RMSD radius. FlowR produced more clusters than Qwen on core (114.4 versus 72.8 per ligand), yet recovered fewer targets (86.2% versus 91.5%). Torsion perturbation likewise produced 74.2 clusters, close to Qwen’s count, but recovered only 78.7% of targets; LoQI reached 86.2% recovery with 20.2 clusters. These contrasting outcomes show why cluster count alone did not provide the same ranking as experimental-conformation recovery (Figure 2B).

The ref cohort showed the same ordering among FlowR, Qwen, and LoQI, whose mean cluster counts were 105.4, 63.7, and 24.4, respectively, and whose recovery rates were 85.5%, 92.5%, and 86.8%. The clustering-radius curves in Figure 2A place these comparisons within a range of geometric resolutions. Because cluster counts depend on both retained ensemble size and clustering order, they describe the spread of the sampled structures rather than the number of biologically accessible states.

Recovery varied with molecular size and flexibility

Stratifying core by heavy-atom count revealed differences that were obscured by the overall recovery rates (Table 2; Figure 3). Qwen, LoQI, and FlowR each recovered all 39 ligands with fewer than 20 heavy atoms, but their performance separated among the 12 ligands with 30–39 heavy atoms: recovery was 66.7% for Qwen, 58.3% for LoQI, 41.7% for ChEMBL3D-PB, and 33.3% for FlowR. Only two core ligands had at least 40 heavy atoms, limiting what could be concluded about the largest molecules.

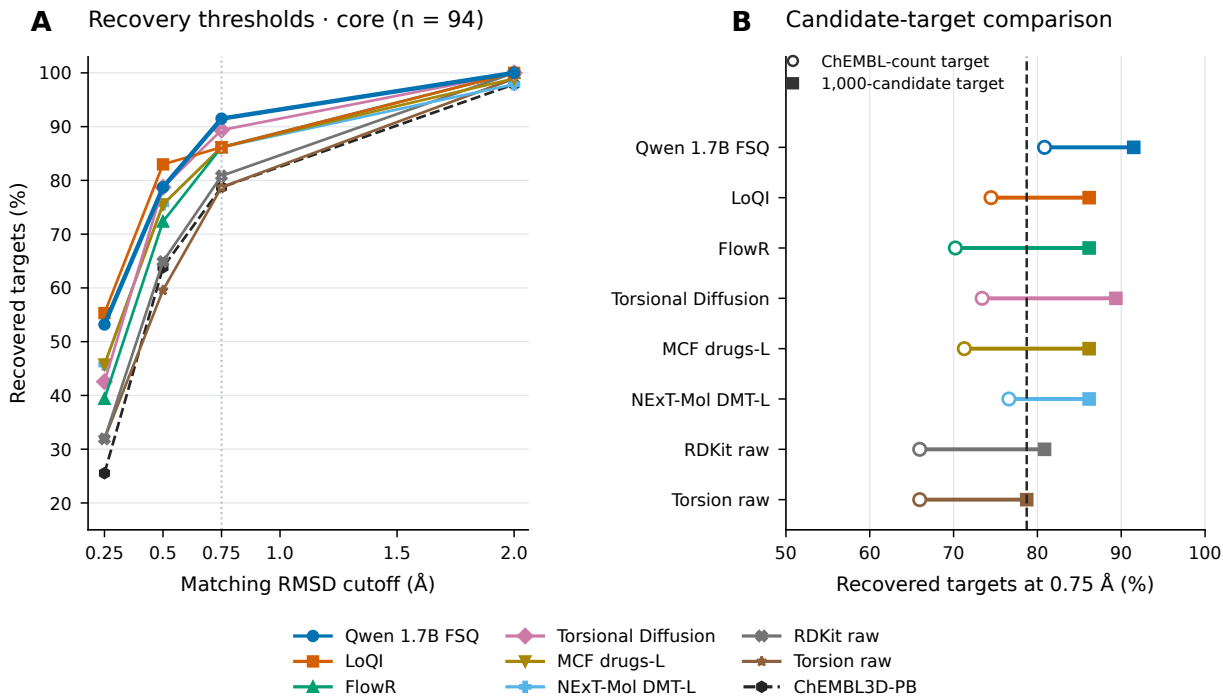


Figure 1: Recovery of experimental ligand conformations in core94. (A) Recovery at the four evaluated matching RMSD cutoffs; the dotted line marks the principal cutoff of 0.75 Å. Points are measured values, with connecting lines as visual guides. (B) Recovery at 0.75 Å for the ChEMBL-count candidate target (open circles) and the 1,000-candidate target (filled squares), followed by PoseBusters filtering. The dashed line is recovery by the finite ChEMBL3D-PB ensemble. Retained counts differ between methods; these are candidate-tier endpoints, not fixed-valid- K curves. All hit rates use the full 94-entry denominator. Qwen is the pretrained 1.7B FSQ model; minimized classical variants are reported in Table 1.

Table 2: Size-stratified recovery for selected contrasting methods at the fixed candidate target. Entries are Hit@0.75 Å percentages over all entries in each group; heavy-atom counts come from a common experimental reference. ChEMBL3D-PB uses its stored pool. The small largest-size groups should be interpreted descriptively.

Cohort	Heavy atoms	Entries	ChEMBL3D-PB	LoQI	FlowR	Qwen
core	< 20	39	94.9	100.0	100.0	100.0
core	20–29	41	78.0	85.4	92.7	92.7
core	30–39	12	41.7	58.3	33.3	66.7
core	≥ 40	2	0.0	0.0	0.0	50.0
ref	< 20	547	93.4	97.8	96.5	99.6
ref	20–29	513	74.9	84.4	90.8	93.8
ref	30–39	152	50.0	64.5	41.4	71.7
ref	≥ 40	24	8.3	29.2	0.0	33.3

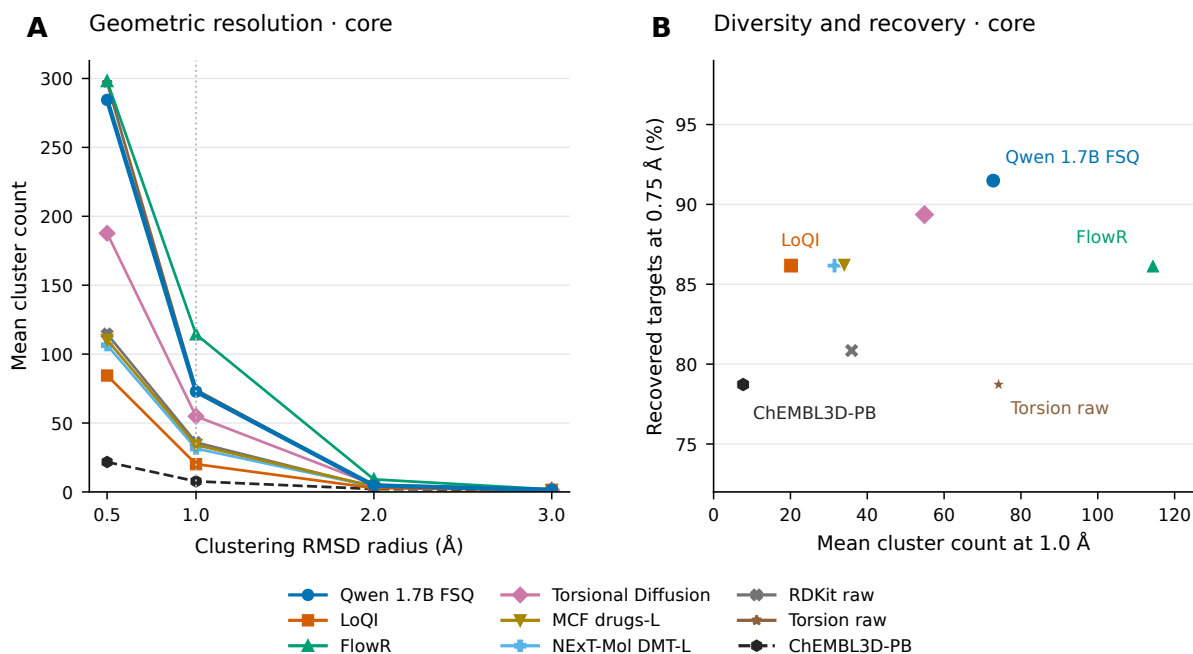


Figure 2: Geometric diversity and experimental-conformation recovery in core94. (A) Mean cluster count at four heavy-atom RMSD clustering radii; the dotted line marks the principal radius of 1.0 Å. (B) Recovery at 0.75 Å versus mean cluster count at 1.0 Å. Generated ensembles use the fixed candidate target after PoseBusters filtering; ChEMBL3D-PB uses its stored pool. Cluster means are conditional on measurable ensembles, whereas recovery includes every mapped entry. Larger cluster counts do not reproduce the recovery ranking. Method colors and symbols match Figure 1A.

The larger ref cohort provided more observations in these size groups and supported the same contrasts. Among its 152 entries with 30–39 heavy atoms, Qwen recovered 71.7%, LoQI 64.5%, ChEMBL3D-PB 50.0%, and FlowR 41.4%. However, recovery remained low among the 24 entries with at least 40 heavy atoms, of which Qwen and LoQI recovered eight and seven, respectively. The size-stratified comparison therefore identified differences between methods without establishing reliable recovery of the largest ligands.

Stratification by flexibility showed a related limitation. Across ref ligands with 0–3, 4–6, 7–8, and at least 9 rotatable bonds, Qwen recovery fell from 99.4% to 94.1%, 73.1%, and 33.3%, while ChEMBL3D-PB recovery was 94.2%, 73.6%, 37.0%, and 17.6% in the same groups. The greatest absolute gain over ChEMBL3D-PB occurred in the 7–8-rotor group rather than among the most flexible ligands. Because size and flexibility are related, these comparisons identify where performance differs without isolating either descriptor as its cause.

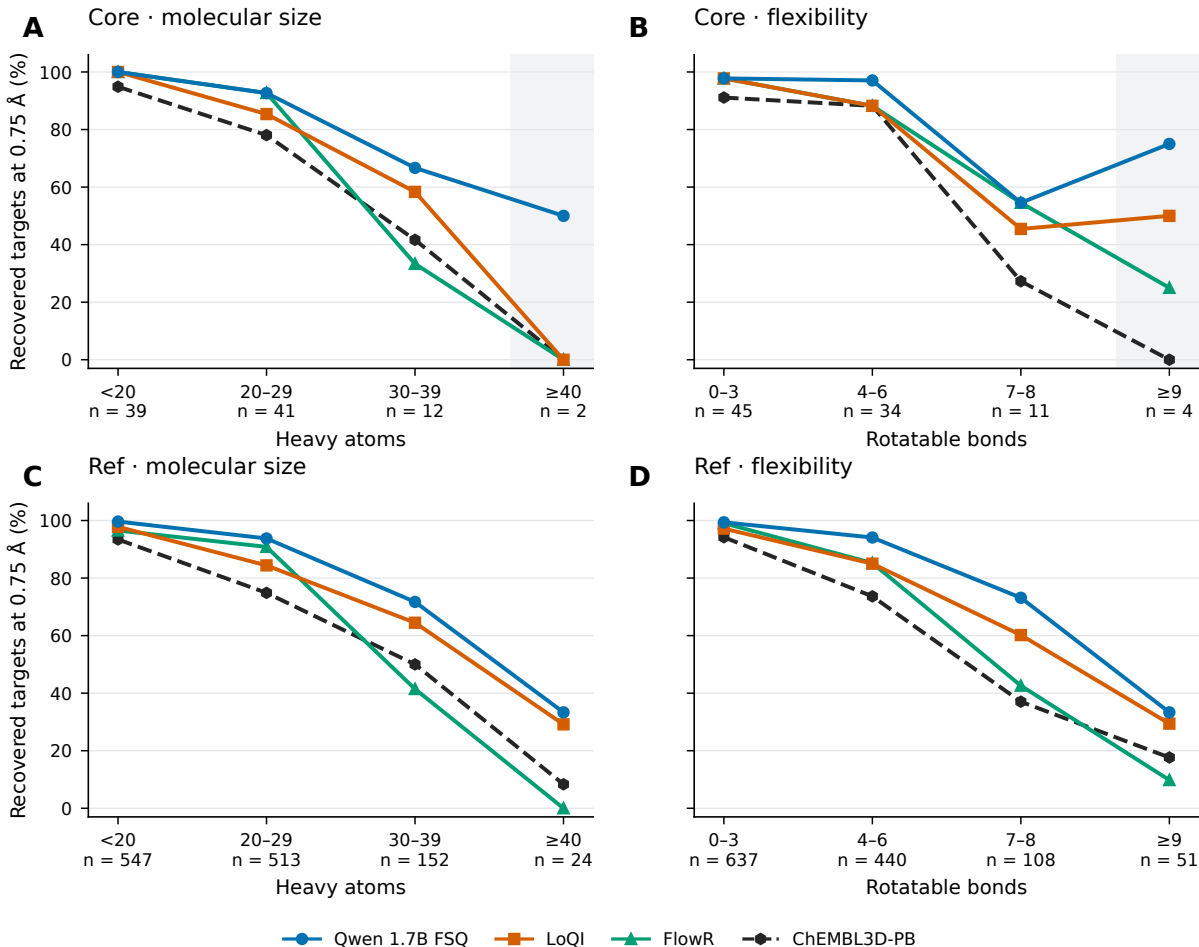


Figure 3: Recovery by molecular size and flexibility for four contrasting methods. Core94 is grouped by heavy-atom count (A) and rotatable-bond count (B); the corresponding ref groups are shown in C and D. Group membership uses the same experimental-reference descriptors for every method. Recovery uses the full group denominator at 0.75 Å. Generated ensembles use the fixed target and ChEMBL3D-PB uses its stored pool. Group sizes are displayed below the axes. Shading identifies the sparsely populated largest-size and highest-flexibility core groups; their estimates are descriptive. Lines connect group estimates as visual guides.

Multiple experimental references separated coverage from concentration

We extended the recovery comparison to a separate panel of 23 drug molecules with 2,450 experimental reference records, allowing us to distinguish coverage of the observed conformations from the concentration of generated samples near them. Coverage was evaluated on the supplied pools and validity was assessed separately; because common PoseBusters filtering and RMSD-failure handling have not yet been harmonized, the comparisons remain descriptive (Methods).

Qwen and LoQI covered similar fractions of the experimental references, with mean COV-R values of 91.0% and 91.2%, respectively (Table 3). Their paired difference of -0.25 percentage points had a wide 95% confidence interval (-9.1 to 8.7), leaving the recall ordering unresolved rather than establishing equivalence. In contrast, LoQI had higher observed-reference precision: COV-P was 59.0%, compared with 46.8% for Qwen, giving a paired Qwen-minus-LoQI difference of -12.2 points (-19.8 to -4.7). Similar reference coverage thus coincided with a greater fraction of LoQI’s generated samples lying near an observed conformation (Figure 4).

Table 3: Multi-reference results for 23 drug molecules. COV-R and COV-P use a strict 0.75 Å cutoff; coverage and matching distances are means over molecules. PB is the separate pooled PoseBusters pass percentage. Coverage is calculated without common PB filtering, and the imported evaluations differ in RMSD-failure handling. These values describe the supplied pools, rather than a harmonized valid-ensemble ranking.

Method	COV-R (%)	COV-P (%)	MAT-R (Å)	MAT-P (Å)	PB (%)
LoQI	91.2	59.0	0.421	0.816	90.9
Torsional Diffusion	78.6	42.6	0.455	1.189	83.4
MCF drugs-L	82.7	39.1	0.481	1.215	63.7
NExT-Mol DMT-L	82.7	36.9	0.464	1.172	62.2
FlowR	81.3	36.6	0.522	1.306	78.5
Qwen 1.7B FSQ	91.0	46.8	0.375	1.000	83.7

The similar mean recall also concealed contrasting molecule-level outcomes. Qwen covered 78.9% of the 57 imatinib references, compared with 5.3% for LoQI, whereas the comparison reversed for the 99 actinonin references (4.0% versus 67.7%). Such differences show why average coverage alone cannot describe performance on individual molecules. They also need to be interpreted relative to the available observations: the reference collection is incomplete, its records need not represent distinct conformational states, and low precision does not establish that unmatched conformers are physically inaccessible.

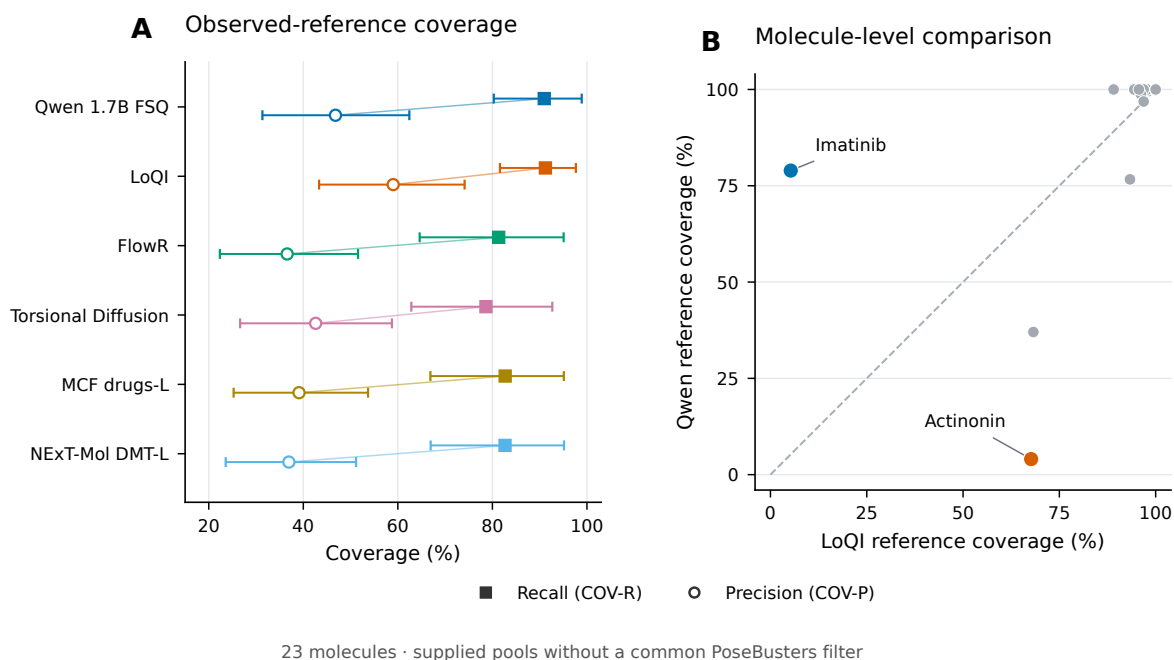


Figure 4: Coverage of multiple experimental conformations for 23 drug molecules. (A) Mean COV-R (filled squares) and COV-P (open circles) at a strict 0.75 Å cutoff. Error bars are 95% percentile intervals from 10,000 molecule bootstrap resamples. (B) Paired molecule-level reference coverage for Qwen 1.7B FSQ and LoQI; the diagonal denotes equal coverage. Imatinib and actinonin illustrate contrasting outcomes. Points can overlap at complete coverage. Metrics describe the supplied pools without common PoseBusters filtering, and RMSD-failure policies differ between imported evaluations; this is not a harmonized valid-ensemble ranking.

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